



Synaptic transistor advances AI and energy-efficient computing

Researchers from Northwestern University, Boston College, and MIT have collaborated and introduced a synaptic transistor inspired by the human brain, capable of higher-level thinking tasks. Their device performs data processing and storage concurrently, resembling the brain's operations. Recent experiments demonstrated its ability to categorise data



An artistic interpretation of brain-like computing (Credit: Xiaodong Yan/Northwestern University)

and perform associative learning. Unlike previous attempts at brain-like computing devices requiring cryogenic temperatures, this synaptic transistor functions at room temperature, operates quickly, consumes minimal energy, and retains information without power. It differs fundamentally from

traditional digital computers where data constantly moves between processors and memory, consuming substantial energy. The team aims to bridge the gap between artificial intelligence (AI) and the human brain. This synaptic transistor uses moiré physics for neuromorphic functionality at room temperature. This could revolutionise AI and energy-efficient computing, shaping the future of technology.

Computational dynamic image sensing with in-sensor informative spikes

Researchers from Hong Kong Polytechnic University, Huazhong University of Science and Technology, and Hong Kong University of Science and Technology have developed a neuromorphic event-based image sensor for efficient dynamic motion capture and processing. Unlike traditional sensors, these new sensors operate uniquely, directly converting dynamic motion into informative spiking signals. These computational event-driven vision sensors can integrate into a spiking neural network for motion recognition. These sensors generate spiking signals with varying amplitudes and polarities, adjusted by electrically programming the photoresponsivity. A noteworthy feature is their non-volatile and multilevel photoresponsivity, mimicking synaptic weights in a neural network. This allows for an in-sensor spiking neural network, effectively merging sensing and computation. It promises faster processing and lower power consumption, a significant advancement in event-based vision sensing and neuromorphic computing.

AI reveals antibiotics for resistant bacteria

MIT researchers harnessed deep learning, a form of AI, to pinpoint compounds capable of eradicating drug-resistant bacteria, including the deadly MRSA. These compounds demonstrated potent MRSA elimination in lab experiments and mouse models, displaying minimal harm to human cells, marking them as promising candidates for drug development.



Using deep learning AI, MIT researchers have discovered a class of compounds that can kill a drug-resistant bacterium that causes more than 10,000 deaths in the US every year (Credit: <https://news.mit.edu>)

A pivotal achievement was understanding the deep-learning model's data types for antibiotic effectiveness prediction, which could enhance the design of more potent drugs. By training the model on an extensive dataset of approximately 39,000 compounds tested against MRSA, they applied the Monte Carlo

tree search algorithm to enhance interpretability, estimating each molecule's antimicrobial potential and offering insights into relevant molecular substructures. By combining toxicity forecasts with antimicrobial assessments, they screened 12 million compounds, ultimately identifying two promising antibiotics against MRSA.

UltraThin lithium anode boosts battery durability and performance

Researchers from Central South University in Changsha, China, have developed a method for creating thin, high-performance lithium anodes for lithium-ion batteries. This improves cycling stability and electrochemical properties, catering to the growing demand for high-quality lithium-ion batteries. Solid-state lithium metal offers a promising alternative to traditional graphite anodes due to its high energy density and capacity. The researchers improved thin lithium metal strips by adding a special zinc additive called dialkyl dithiophosphate (ZDDP). ZDDP enhanced hardness at the interface, prevented structural deterioration, regulated lithium deposition during plating and stripping processes, and enabled faster plating and stripping compared to other materials. The enhanced electrochemical properties were attributed to the formation of a robust artificial solid electrolyte interface (SEI) layer with a strong affinity for lithium. This pioneering effort in developing ultra-thin lithium for solid-state batteries involved a production process using a zinc-based oil, similar to rolling dough through a pasta machine.